GENERIC CELLULAR CONTEXTS IN CUBICAL TYPE THEORY REALIZE THEIR CUBE PRESHEAVES ON THE NOSE

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ABSTRACT. A companion paper showed that in cubical type theory with a De Morgan interval, a generic higher *loop* spans, under definitional equality, a boundary-collapsed representable presheaf on the De Morgan cube category — a "De Morgan sphere" — and left open what happens for cells with *generic* boundaries. This paper answers that question. For a cellular context — a cubical computad: generic vertices, generic edges between them, generic squares with prescribed edges, and so on — the cubes definable by pure reparametrization form, modulo definitional equality and naturally in the cube dimension, exactly the *cellular presheaf* of the computad: the colimit of one representable per cell, glued along the attachment maps. The gluing is definitional in the strongest sense: each face of a generic cell *is* its prescribed boundary cell on the nose, and no further identifications occur — in particular, cells sharing a face are identified along exactly that face. For a single cell with generic boundary the layer is the full representable, with no collapse; the spheres of the companion paper are recovered as the boundary-degenerate limit. As a by-product we isolate a phenomenon absent from the Kan world: *strict fillers are not unique* — a boundary in the strict layer can admit several distinct fillers, reflecting the non-injectivity of the restriction of free De Morgan polynomials to the faces of a cube. All positive and negative instances are machine-checked in a self-contained proof-assistant kernel, and every statement is a decidable, assumption-free fact about the algorithmic equality of a minimal fragment.

1. INTRODUCTION

Cubical type theories [1] equip the identity structure of types with definitional operations indexed by a De Morgan algebra: reparametrizing a cell along $\wedge$, $\vee$, $\neg$ — the connections and reversals — happens on the nose, not up to higher cells. A companion paper [7] determined the exact extent of this strictness over contexts of generic *loops*: the definable cubes form a wedge of *De Morgan spheres*, boundary-collapsed representable presheaves $\square_{\mathrm{DM}}(-,[n])/\partial$ on the De Morgan cube category $\square_{\mathrm{DM}}$, one per loop, and nothing else is identified. The collapse of the boundary was forced by the

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choice of generic objects: a loop's faces are reflexivities, and reflexivities absorb everything applied to them.

That paper left open the *unpointed* question: what does a generic cell with a *generic* boundary span? The present paper answers it, and the answer upgrades the picture from wedges of quotients to genuine cell complexes.

Cellular contexts. By a *cellular context* (a *cubical computad*) we mean a context of generic cells declared dimension by dimension: generic points $a, b, \cdots : A$; generic edges between prescribed points, $p : \mathsf{Path}\, A\, a\, b$; generic squares with four prescribed edges, $Q : \mathsf{PathP}\, (\langle i \rangle\, \mathsf{Path}\, A\, (r\, i)\, (s\, i))\, p\, q$; and so on. This is the type-theoretic incarnation of a computad (polygraph) [4, 5] over the cube category: free generators in each dimension, attached along specified lower cells. Every computad C has a *cellular presheaf* $|C|$ on $\square_{\mathrm{DM}}$ — one representable $\square_{\mathrm{DM}}(-, [n_c])$ per cell c, glued along the attachment maps — the cubical analogue of the cell complex a CW-presentation describes.

Main results. Let $\equiv$ denote the definitional (algorithmic) equality of the calculus (Section 3).

(1) *Attachment is definitional* (Section 4). Each face of a generic cell computes, on the nose, to its prescribed boundary cell: $\langle j \rangle\, Q(0)(j) \equiv p$, $\langle i \rangle\, Q(i)(0) \equiv r$, and so on; corners compute through two steps to the prescribed vertices. The structure maps of the cellular presheaf are thus judgmental equalities, not merely admissible ones.

(2) *The realization theorem* (Sections 5–6). The pure-reparametrization cubes over a cellular context, modulo $\equiv$ and naturally in the cube dimension $[m]$, are exactly the cellular presheaf:

$$\{m\text{-cubes}\}/\!\equiv \;\cong\; |C|([m]).$$

For a single cell with generic boundary this is the *full* representable $\square_{\mathrm{DM}}(-, [n])$ — no collapse; for cells sharing a face, the gluing identifies exactly the shared face (machine-checked in both directions: the cross-cell face equation holds definitionally, and interior cubes of distinct cells with *identical boundaries* remain separated).

(3) *Strict fillers are not unique* (Section 7). The elements $i \wedge \neg i$ and $(i \wedge \neg i) \wedge (j \vee \neg j)$ of the free De Morgan algebra are distinct, yet restrict equally to all four faces of the square. Consequently two *different* strict squares can fill the *same* boundary. Where Kan filling is unique up to homotopy, strict filling is non-unique on the nose — for the entirely algebraic reason that restriction of free De Morgan polynomials to the boundary of a cube is not injective.

(4) *Sphere recovery* (Section 8). Degenerating the boundary of the computad to a point recovers the wedge-of-spheres theorem of [7]: the collapse $\square_{\mathrm{DM}}(-, [n]) \twoheadrightarrow \square_{\mathrm{DM}}(-, [n])/\partial$ is the image, under realization, of collapsing the boundary cells.

All positive and negative instances are elaboration-time checks in the artifact kernel (Section 9); by the decidability theorem imported from the first companion paper [6], each is a decided, assumption-free fact.
Honest positioning. As in [7], the positive direction — that the De Morgan laws and the attachment equations hold — is by design of the theory. The content is the exactness: the identifications are *exactly* those of the cellular presheaf, no more (the separations, including the delicate same-boundary ones behind non-uniqueness of fillers) and no less (the attachment equations). We know of no prior statement of the realization theorem or of the non-uniqueness phenomenon. The proofs are elementary given two small kernel facts (K1, K2 below) and are designed to be checked against the artifact.
Relation to the companions. This paper is the third in a series but is written to be read independently. From [6] we import only the decidability of the algorithmic equality on the minimal fragment (K3) and the two kernel facts K1, K2; from [7] we import the De Morgan cube category and the sphere theorem, which the present results generalize and recover. Nothing here depends on the transport-layer analysis of [6] (Remark 3.1).

2. The De Morgan cube category, computads, and cellular presheaves

Definition 2.1 ([7, Def. 2.2])**.** $\square_{\mathrm{DM}}$ has objects $[m]$, $m \geq 0$, and morphisms $\square_{\mathrm{DM}}([m],[n]) := \mathrm{DM}(i_1,\dots,i_m)^n$, tuples of free De Morgan polynomials, composed by substitution; the identity of $[n]$ is the generator tuple. $\mathrm{DM}(\vec{i})$ is the free De Morgan algebra (no excluded middle), with canonical antichain disjunctive normal forms; $0,1$ are its *endpoints*. The *k-th face* is $\delta^\varepsilon_k := (i_1,\dots,\varepsilon,\dots,i_{n-1}) : [n-1] \to [n]$; a morphism is *face-factoring* if it factors through some δ^ε_k, equivalently iff some coordinate is an endpoint *of the free algebra*.

Definition 2.2 (Cellular contexts)**.** A *cellular context of dimension* ≤ 2 over a base type $A : \mathcal{U}$ is a context declaring, in order:

- generic vertices $a_1, \dots : A$;
- generic edges $e : \mathsf{Path}\,A\,a\,a'$ between prescribed vertices;
- generic squares $Q : \mathsf{PathP}\,(\langle i\rangle\,\mathsf{Path}\,A\,(r\,i)\,(s\,i))\,p\,q$ with four prescribed edges p,q,r,s whose endpoints match at the corners.

Higher cells are declared analogously along the cube-boundary telescopes; we spell out dimension ≤ 2, which is where our machine evidence lives, and treat higher dimensions uniformly (Remark 5.3). We call the data a *cubical computad* C; it is precisely a polygraph [4, 5] over $\square_{\mathrm{DM}}$ presented type-theoretically.

Definition 2.3 (Cellular presheaf)**.** The *cellular presheaf* $|C|$ of a computad C is the presheaf on $\square_{\mathrm{DM}}$ with

$$|C|([m]) \;:=\; \Big(\coprod_{c\in C} \square_{\mathrm{DM}}([m],[n_c])\Big)\Big/\sim,$$

where $\sim$ is generated by the attachments: if $\vec{\varphi} = \delta_k^{\varepsilon} \circ \vec{\chi}$ face-factors, then $(c,\vec{\varphi}) \sim (\partial_k^{\varepsilon} c, \vec{\chi})$, for $\partial_k^{\varepsilon} c$ the prescribed boundary cell. Equivalently, $|C|$ is the colimit of the representables $\square_{\mathrm{DM}}(-,[n_c])$ along the face inclusions specified by the attachments — the cubical cell complex presented by C.

3. The calculus and imported facts

We work in the reparametrization fragment of a CCHM-style cubical type theory [1]: path types over a base type, path abstraction and application; Kan operations are excluded from the layer under study. Definitional equality is the algorithmic equality $\equiv$ of the concrete normalization-by-evaluation kernel of [6], whose two load-bearing clauses are:

(K1) *Endpoint collapse is normal-form exact*: applying a path value at $r \in \mathrm{DM}(\vec{i})$ collapses to the stored endpoint iff r is an endpoint of the free algebra (the kernel decides by antichain normal forms). For a value of a *dependent* path type $\mathsf{PathP}\,(\langle i\rangle T)\,l\,u$, the stored endpoints are l, u; for a neutral spine, the stored endpoint *annotations* of its type.

(K2) *Conversion of neutrals is spine-wise*: same head variable, same length, convertible endpoint annotations, interval arguments equal in the free algebra.

(K3) *Decidability, assumption-free* [6]: on the minimal fragment, evaluation and conversion terminate and $\equiv$ is decidable.

Remark 3.1 (Robustness)**.** The fragment involves no transport, so all results are insensitive to the transport-layer design of [6] and hold verbatim for a standard CCHM-style conversion; we use the kernel of [6] because there they are unconditional decidable facts (K3).

Throughout, fix a cellular context; for a generic square Q with edges p, q, r, s (as in Definition 2.2) and $\varphi, \psi \in \mathrm{DM}(i,j)$ write

$$Q(\varphi)(\psi)$$

for the doubly applied cell (with the boundary-typed endpoint annotations) and $A_Q(\vec{\varphi}) := \langle \vec{i}\,\rangle\, Q(\varphi_1)(\varphi_2)$ for the reparametrized cube, and likewise for edges.

4. Attachment is definitional

Theorem 4.1 (Attachment equations)**.** *For a generic square Q with prescribed edges p, q, r, s:*

$$\langle j\rangle\, Q(0)(j) \equiv p, \quad \langle j\rangle\, Q(1)(j) \equiv q, \quad \langle i\rangle\, Q(i)(0) \equiv r, \quad \langle i\rangle\, Q(i)(1) \equiv s,$$

and corners compute through two steps: $\langle i\rangle\langle j\rangle\, Q(0)(0) \equiv \langle i\rangle\langle j\rangle\, a_{00}$*, etc. More generally the whole face square is the edge square:* $\langle i\rangle\langle j\rangle\, Q(0)(j) \equiv \langle i\rangle\langle j\rangle\, p(j)$*.*

Proof. By K1 at the outer application: $Q(0)$ collapses to the stored endpoint of the PathP type, which is the *prescribed edge* p — a generic cell, not a reflexivity; the remaining application and η give the first equation. For the third, $Q(\varphi)$ at non-endpoint φ is a neutral spine whose stored endpoint annotations are $r(\varphi), s(\varphi)$; K1 at the inner application returns them, and η finishes. Corners iterate the two collapses. Machine probes: `gbAttPD`, `gbAttQD`, `gbAttRD`, `gbAttSD`, `gbCornerD`, `gbEdgeSqD`. □

Remark 4.2. This is the exact point at which the generic-boundary situation departs from the loops of [7]: there the stored endpoints were reflexivity towers, through which every further application computed to the base point — producing the $/\partial$ collapse. Here the stored endpoints are generic cells, and the face-factoring morphisms land in the boundary's own layer instead of a single point. The De Morgan laws themselves remain strict in the presence of a generic boundary: the unit and absorption laws hold at the full dependent square type (machine probes `gbEtaD`, `gbAbsorbD`, `gbAbsorbPerturbD`).

5. The single-cell realization

Lemma 5.1 (Value computation, generic boundary)**.** *Over the square computad (four vertices, four edges, one square), open the m binders of a definable cube at generic dimensions. The body evaluates to exactly one of:*

(1) *a vertex variable;*
(2) *an edge spine $e(\chi)$ with $e \in \{p, q, r, s\}$ and χ a non-endpoint;*
(3) *a full spine $Q(\varphi)(\psi)$ with both coordinates non-endpoints.*

A tuple (φ, ψ) face-factoring as $\delta^{\varepsilon} \circ \chi$ evaluates to the same value *as the corresponding boundary composite: the edge spine of the prescribed face at χ (or a vertex, when χ is again an endpoint).*

Proof. K1 case analysis as in Theorem 4.1; the three cases are exhaustive because collapse consumes coordinates until either none remain (vertex) or the first non-endpoint argument freezes the spine. □

Theorem 5.2 (Single cell, full representable)**.** *The pure-reparametrization m-cubes over the square computad, modulo $\equiv$, are naturally isomorphic to $|C| = \square_{\mathrm{DM}}(-, [2])$ — the* full *representable, the isomorphism sending $\vec{\varphi}$ to $A_Q(\vec{\varphi})$ and the face-factoring morphisms to the boundary composites.*

Proof. Surjectivity is Lemma 5.1: every value is a vertex, edge, or full spine, i.e. the image of a morphism (for the square computad, every boundary composite is a face-factoring morphism of $[2]$, so $|C|([m]) = \square_{\mathrm{DM}}([m], [2])$ — attachments here *create* no quotient, they only relate the coproduct's summands, and every summand embeds by face composition).

Well-definedness (the presheaf identifications hold) is Theorem 4.1.

Injectivity: two full spines are convertible iff their tuples agree (K2, interval arguments componentwise in the free algebra); two edge spines iff same edge and same argument (K2: the four edges are distinct head variables); vertex values are the four distinct vertex variables; and the three classes are mutually non-convertible (distinct heads; a variable is not a spine). A face-factoring tuple and a non-face-factoring tuple land in different classes by Lemma 5.1. Finally two face-factoring tuples with the same value have the same boundary composite, hence are equal in $\square_{\mathrm{DM}}([m],[2])$ — e.g. $p(\chi) = p(\chi')$ forces $\chi = \chi'$, and cross-edge coincidences occur only at corners, where the tuples agree. $\square$

Remark 5.3 (Higher dimensions)**.** For a generic n-cell attached along the full cube-boundary computad of dimension n, Lemma 5.1 and its consequences go through verbatim by induction on n: K1 peels endpoint coordinates one at a time, each collapse landing in the boundary computad one dimension down, and K2 separates the resulting spines by head and length. We have carried out the machine verification for $n \leq 2$; the induction step involves no new phenomenon, and we state the general case as a theorem with this uniform proof.

6. Gluing, and the main theorem

The induction step for several cells is the two-square computad: squares Q_1 (edges p, m, r_1, s_1) and Q_2 (edges m, q, r_2, s_2) *sharing* the edge m.

Theorem 6.1 (Gluing along a shared face)**.** (1) *(Identification.) The two faces agree across the two cells, definitionally:*

$$\langle i\rangle\langle j\rangle\, Q_1(1)(j) \;\equiv\; \langle i\rangle\langle j\rangle\, Q_2(0)(j) \qquad (\textit{both} \equiv \textit{the edge square of } m).$$

(2) *(No over-identification.) Let* $E_0 := i \vee \neg i \vee j \vee \neg j$ *and* $D_0 := i \wedge \neg i \wedge j \wedge \neg j$ *— the edge-one and edge-zero elements of* $\mathrm{DM}(i,j)$*. The squares*

$$S_1 := \langle i\rangle\langle j\rangle\, Q_1(E_0)(j), \qquad S_2 := \langle i\rangle\langle j\rangle\, Q_2(D_0)(j)$$

have their entire boundaries *on m: together with the edge square* $\langle i\rangle\langle j\rangle\, m(j)$*, both inhabit the type* $\mathsf{Path}\,(\mathsf{Path}\,A\,a_{10}\,a_{11})\,m\,m$*. All three are pairwise non-convertible.*

Proof. (1) Both sides collapse by K1 to the shared cell m (Theorem 4.1 applied in each square), hence to each other. Machine probes `seAttM1D`, `seAttM2D`, `seGlueD`. (2) E_0 and D_0 are non-endpoints of the free algebra whose restrictions to all four faces are 1 resp. 0; so S_1, S_2 are full spines with heads Q_1 resp. Q_2, the edge square is an m-spine, and K2 separates the three by head. Well-typedness at the common type is the boundary computation of (1). Machine probes `seS1CtrlD`, `seS2CtrlD`, `seSmCtrlD` and the three separation guards. $\square$

Theorem 6.2 (Realization)**.** *For every cubical computad C (of dimension ≤ 2; higher dimensions per Remark 5.3), the pure-reparametrization m-cubes over the context of C, modulo the algorithmic equality, form naturally in $[m]$ the cellular presheaf:*

$$\{m\text{-}cubes\}/\!\equiv\ \cong\ |C|([m]).$$

All of this is decidable and assumption-free by K3.

Proof. Induction on the cells of C. Adding a vertex or an edge is the base analysis of Lemma 5.1 (an edge over its two vertices is the one-dimensional instance of Theorem 5.2). Adding a square Q attached along prescribed edges: the new definable bodies are exactly the Q-spines (Lemma 5.1, whose proof is local to the cell and its boundary); the attachment identifications hold definitionally (Theorem 4.1), matching the presheaf's $\sim$; and no new identifications occur — between a Q-spine and anything previously definable (K2, head separation; the same-boundary cases are exactly those of Theorem 6.1(2), which is the general no-merge computation: the boundary-collapsed interior of a new cell never merges with old material or with another cell's interior). Naturality in $[m]$ — compatibility with reparametrization — is the strict functoriality of the actions, proved in [7, Lemmas 4.2–4.3] and re-checked at dependent square types here (`gbEtaD`, `gbAbsorbD`). □

7. Strict fillers are not unique

In the Kan world, a boundary determines its fillers up to homotopy. In the strict layer the situation is sharper in one way — fillers are literal — and looser in another:

Theorem 7.1 (Non-uniqueness of strict fillers)**.** *Over the square computad there exist two non-convertible squares with the* same *boundary: with $\varphi_1 := i \wedge \neg i$ and $\varphi_2 := (i \wedge \neg i) \wedge (j \vee \neg j)$,*

$$\langle i\rangle\langle j\rangle\, Q(\varphi_1)(j)\ \not\equiv\ \langle i\rangle\langle j\rangle\, Q(\varphi_2)(j),$$

yet both inhabit $\mathsf{PathP}\,(\langle i\rangle\,\mathsf{Path}\,A\,(r(i\wedge\neg i))\,(s(i\wedge\neg i)))\,p\,p$.

Proof. $\varphi_1 \neq \varphi_2$ in the free algebra: absorption fails because the clause $\{i, \neg i\}$ contains neither $\{j\}$ nor $\{\neg j\}$, so $(i \wedge \neg i) \wedge (j \vee \neg j) \neq i \wedge \neg i$. Yet their restrictions to the four faces coincide (0 on the i-faces, $i \wedge \neg i$ on the j-faces), so the two squares have convertible boundaries; being non-degenerate full spines with distinct interval arguments, K2 separates them. Machine probes `gbFill1D`, `gbFill2D` and the separation guard. □

Proposition 7.2 (The minimal perturbation is absorbed)**.** *The edge-vanishing elements of* $\mathrm{DM}(i,j)$ *are exactly* 0 *and $D_0 = i \wedge \neg i \wedge j \wedge \neg j$, and $D_0 \leq \varphi$ for* every $\varphi \neq 0$ *(the clause of D_0 contains every literal, hence contains every clause). Consequently join-perturbation by D_0 never produces a new filler: $\varphi \vee D_0 = \varphi$ (machine probe `gbAbsorbPerturbD`). The non-uniqueness of Theorem 7.1 therefore arises only through meet-constructions such as φ_2,* *i.e. through the non-injectivity of the boundary-restriction map* $\mathrm{DM}(i,j) \to \mathrm{DM}(j)^2 \times \mathrm{DM}(i)^2$ *— an intrinsic feature of free De Morgan algebras, faithfully reflected by definitional equality.*

Remark 7.3. Theorem 7.1 is consistent with the realization theorem — the representable $\square_{\mathrm{DM}}(-,[2])$ genuinely has several elements over a single boundary configuration — but it is worth isolating because it contradicts a natural first guess (that strict fillers, when they exist, are unique) and because it gives the boundary-restriction non-injectivity of free De Morgan algebras a geometric meaning inside type theory.

8. Recovering the spheres

The companion's generic loops are the boundary-degenerate limit of the present setting: substituting the constant computad (one vertex, reflexivity boundaries) for the boundary cells sends the square computad to the 2-loop context of [7]. Under this substitution the stored endpoints of the spine collapse to reflexivity towers, through which K1 computes to the base point — precisely the mechanism that turns the full representable into the boundary-collapsed one:

Proposition 8.1. *Boundary degeneration realizes the quotient: the substitution of the point computad for the boundary carries the isomorphism of Theorem 5.2 to the sphere isomorphism* $\{cubes\}/{\equiv} \cong \square_{\mathrm{DM}}(-,[2])/\partial$ *of* [7]*, the induced map on presheaves being the canonical collapse* $\square_{\mathrm{DM}}(-,[2]) \twoheadrightarrow \square_{\mathrm{DM}}(-,[2])/\partial$*.*

Proof. The substitution sends attachment values (edge spines, vertices) to reflexivity towers and the base point; Lemma 5.1 degenerates clause-wise to the value computation of [7, Lemma 5.1], identifying exactly the face-factoring classes. $\square$

Together, Theorems 5.2, 6.2 and Proposition 8.1 give a uniform picture: *the definitional equality of cubical type theory realizes cubical computads as their cell complexes, on the nose; collapsing boundaries collapses realizations accordingly.*

9. Machine verification

All instances quoted above are elaboration-time checks in the self-contained Lean 4 cubical kernel of [6]: repository

`https://github.com/r-shibusawa/cubical-strict-j`.

The probes of this paper are the files `LibGenBoundary.lean` (attachments, corners, edge squares, unit and absorption at dependent square types, filler non-uniqueness) and `LibSharedEdge.lean` (cross-cell gluing and the no-merge separations), with the supporting proof notes `docs/GenericBoundary.md`. They are included from release `v1.3.0`, archived at DOI `10.5281/zenodo.TODOD`. A successful `lake build` replays every check; by K3 each check is a decision. As in the companions, the kernel supplies machine-checked *instances*; the theorems are pen-and-paper results about the conversion algorithm, whose two load-bearing clauses (K1, K2) are small inspectable functions.

10. Related work and outlook

Computads/polygraphs as presentations of free higher structures go back to Street and Burroni [4]; see [5] for a modern account. The present results interpret the *free* structure they present inside cubical type theory [1] literally: definitional equality itself realizes the presented cell complex over the De Morgan cube category [2, 3], with nothing added and nothing collapsed. The companion papers [6, 7] supply the metatheoretic platform (decidability of the algorithmic equality on the fragment) and the pointed special case (loops and spheres).

Two directions seem worthwhile. First, *mixed layers*: our fragment excludes the Kan operations; over a cellular context, the full theory's cubes include composites and fillers, and the inclusion of the strict layer into the full layer is a type-theoretic incarnation of the unit "free strict $\to$ free weak" comparison, whose exact structure (in particular which weak cells are detected by strict boundaries, given Theorem 7.1) deserves study. Second, *presentations*: since realization is exact, one can read off normal forms for the free cubical ω-category on a computad in the De Morgan signature from the type theory's normal forms — suggesting a proof-assistant-based approach to word problems for cubical polygraphs.

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